

NEW RICHMOND, WI, USA

WMO: 722179

Lat: 45.150N

Lon: 92.533W

Elev: 304

StdP: 97.73

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04968

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-22.8	-29.8	0.2	-24.9	-27.4	0.3	-22.5	10.8	-9.3	9.4	-9.0	1.9	300	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	31.1	22.1	29.1	21.1	27.6	20.2	23.9	28.6	22.8	27.2	21.8	25.7	5.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	17.9	26.2	21.4	16.7	25.0	20.5	15.8	24.1	73.2	28.7	69.1	27.6	65.2	25.7	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	8.5	7.7	-29.9	33.9	2.7	1.8	-31.9	35.2	-33.5	36.2	-35.0	37.2	-37.0	38.5		
			-30.1	25.8	2.7	1.3	-32.0	26.7	-33.6	27.5	-35.1	28.2	-37.1	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-9.7	-8.5	-0.6	6.7	13.6	18.8	21.3	19.8	16.2	8.6	1.1	-6.8
	DBStd	12.19	7.29	6.69	7.00	5.47	4.94	3.30	3.06	3.12	4.43	4.95	5.67	6.47
	HDD10.0	2499	612	518	336	127	20	0	0	0	6	88	272	521
	HDD18.3	4477	870	752	588	350	163	33	8	20	91	305	518	779
	CDD10.0	1318	0	0	7	27	130	263	349	303	192	44	4	0
	CDD18.3	254	0	0	0	1	15	46	99	64	27	3	0	0
	CDH23.3	2182	0	0	2	8	159	374	886	521	213	20	0	0
	CDH26.7	542	0	0	0	0	42	82	251	122	45	1	0	0
Wind	WSAvg	3.3	3.5	3.5	3.5	4.0	3.7	3.1	2.6	2.5	3.0	3.3	3.6	3.5
Precipitation	PrecAvg	840	23	20	43	73	100	122	112	106	92	77	45	26
	PrecMax	1088	57	43	84	191	196	209	218	197	196	164	128	57
	PrecMin	638	5	3	13	18	25	47	56	23	24	26	1	7
	PrecStd	118	14	10	19	38	46	40	44	47	43	45	37	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	9.9	21.4	24.6	31.1	31.7	33.7	32.1	30.7	26.1	18.9	7.9
		MCWB	2.5	7.7	14.6	14.6	19.2	21.9	23.5	22.2	20.8	17.8	12.3	5.9
	2%	DB	2.7	4.8	16.1	21.1	27.2	29.0	31.8	29.8	27.7	22.7	14.8	5.0
		MCWB	1.3	2.3	11.2	12.5	17.8	20.3	23.5	21.6	20.4	16.0	10.2	3.2
	5%	DB	1.3	2.6	12.3	18.8	24.9	27.1	29.3	27.8	26.0	19.9	12.0	2.7
		MCWB	0.5	0.8	8.3	10.6	16.2	19.1	21.9	20.2	19.0	14.7	8.2	1.5
	10%	DB	0.1	0.6	8.2	16.1	22.3	25.6	27.6	26.3	23.8	16.6	9.2	1.3
		MCWB	-0.5	-0.7	5.2	9.2	14.9	18.1	20.8	19.4	18.3	11.8	6.2	0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	6.7	15.8	15.6	22.2	24.0	26.3	24.6	23.7	20.3	13.8	7.5
		MCDB	4.4	9.7	20.2	21.0	28.9	29.6	29.6	29.0	27.2	22.8	17.2	7.6
	2%	WB	1.6	2.7	12.3	13.7	19.6	22.1	24.5	23.2	22.0	17.4	11.0	3.2
		MCDB	2.7	4.4	15.2	19.7	24.4	26.6	29.3	27.5	25.7	21.7	14.5	4.5
	5%	WB	0.6	0.9	8.3	11.8	17.9	20.7	23.2	22.2	20.7	14.7	8.2	1.8
		MCDB	1.2	2.4	12.3	16.7	22.2	24.8	27.8	25.5	23.8	18.6	10.9	2.7
	10%	WB	-0.5	-0.4	5.5	10.2	16.4	19.6	22.1	21.1	19.4	12.7	6.4	0.7
		MCDB	0.1	0.5	8.3	15.4	20.6	23.7	26.4	24.3	22.9	15.9	9.3	1.1
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.9	9.7	10.6	11.3	10.5	11.2	11.1	11.1	9.8	8.2	7.6
		MCDBR	8.1	9.9	12.9	15.5	14.5	12.5	12.6	12.5	12.7	13.7	12.1	8.2
	5% WB	MCWBR	7.0	8.0	7.9	8.5	6.8	5.9	5.9	5.6	6.5	7.8	8.0	7.0
		MCWBR	7.7	9.3	11.9	13.4	11.5	10.4	11.0	10.4	10.2	11.1	11.0	7.1
Clear-Sky Solar Irradiance	taub	taub	0.258	0.281	0.318	0.367	0.400	0.392	0.384	0.395	0.366	0.319	0.286	0.264
		taud	2.399	2.323	2.345	2.309	2.264	2.323	2.369	2.347	2.401	2.519	2.508	2.422
	Ebn at Noon	Ebn at Noon	872	907	914	892	870	876	879	855	853	857	832	824
		Edh at Noon	71	94	107	122	132	125	118	116	101	78	65	64
All-Sky Solar Radiation	RadAvg	RadAvg	1.71	2.80	3.86	4.61	5.32	5.91	6.25	5.22	4.06	2.56	1.63	1.19
	RadStd	RadStd	0.19	0.28	0.34	0.46	0.43	0.45	0.21	0.37	0.28	0.34	0.15	0.14

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
Station Only	N/A	-1.87	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page